# SIMATS ENGINEERING
## ASSESSMENT TOOL 3

**COURSE NAME:** Artificial Intelligence **COURSE CODE:** CSA17

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**COURSE OUTCOME COVERED**

***CO1:*** Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

*Assessment Tool Weightage: 30%*

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**QUESTIONS: 2**

**Duration : 45 Minutes**
**Total Marks:20**

Annexure A

**Descriptive Written Test**

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***Code of Conduct:***

*I* K.FLORINA (192521479) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
*I understand that any violation of academic integrity rules will result in disciplinary action.*

***Signature of the student:***
K. Florina

K. FLORINA
192521479

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1. Intelligent agents in a smart home automation System.

## Introduction:

An intelligent agent is a system that perceives its environment through sensors and takes action through actuation to achieve some objectives.

In a smart home, sensors detect motion, temperature, light intensity while actuation control lights, air conditioners and alarms.

1. **Simple reflex agent:**

A simple reflex agent selects actions based only on the current percept using condition-action rules.

**Advantages:**

* simple to design and implement.
* Responds quickly to real time inputs.

Disadvantages :

* Does not remember past events.
* cannot learn user preference.

2. Model based agent :

A model based agent maintains an internal state of the environment using current input and previous observations.

Advantages :

* Handles partially observable environments.
* uses past information to make better decisions.

Disadvantages :

* Require memory and a model of the environment.
* more complex than a simple reflex agent.

3. Goal based Agent :

A goal based agent selects actions that help achive a specific goal.

Eg in smart home :

*Achieve a comfortable room temperature for 24°C.

Advantages :

*Achieve a comfortable room temperature of 24°C.

*can choose different actions to achieve the same goal.

Disadvantages :

*Requires searching or planning to select action.

*May consume more computational resources.

4. utility - Based agent :

A utility based agent evaluates possible actions using a utility function that measures how each outcomes.

**Advantages:**

↳ Balances multiple computing objectives.

↳ Supports trade-off between Comfort, cost.

**Disadvantages:**

↳ Designing a suitable utility function is difficult.

↳ Requires more computation and information.

↳ Incorrect utility weights can produce undesirable decision.

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2. Uninformed search strategies for a robot in maze.

1. **uniform cost search:**

* uniform cost search expand the node with the lowest to save path cost, $g(n)$ from the starting position.

* It uses priority que to select the least cost node.

## Working Procedure :

1. place the starting position in the priority queue with cost 0.
2. remove the node with the lowest path cost.
3. check whether the node is exit.
4. If not exit, explore universal neighbouring positions.
5. calculate the cumalative cost.
6. continue until the exit found.

## Time and space complexity.

Time : $O(b^{l} + LC * l\varepsilon))$

space : $O(b^{l} + LC * l\varepsilon))$

2. Ituative depending search (IDS)

Ituative depending search repeatedly perform depth limited first search, including the depth limit at each ituation.

## Working Procedure :

1. Begin with a depth limit of 0.
2. Search the maze upto the current depth limit.
3. If they increase or not found. increase the depth limit by,
4. Repeat the space from the starting position.
5. stop the when search.

Eg: IDS seaches depth 0, then depth 1, then depth 2.

## Advantages :

- uses much less memory than VCS.
- Complete when the branching factor is finite.
- Finds a shallowest solution when each step has equal cost.
- useful when the solution depth is unknown.

## Disadvantages :

- Repeats the exploration of upper level nodes.
- May take longer if they exist in very depth.
- Does not guarenty a minimum - cost path when step cost differs.

## Time and space complexity :

For branching factor b and shallowed solution depth d.

Time : $O(b^d)$

space : $O(b^d)$